
BY 521P: Computational Biology and Cellular Bioprocess Technology Lab

Course Code	BY 521P
Course Name	Computational Biology and Cellular Bioprocess Technology Lab
L-T-P-C	0-0-2-1
Prerequisites	None
Intended for	M. Tech. Biotechnology
Distribution	Core Lab for M. Tech. Biotechnology
Approval	9th Senate (May 2016)

Course Contents

- Statistical analysis using biological data using statistical software (R or excel)
- Basic scripting – Perl
- Biological databases and sequence file formats
- Local alignment + global alignment exercise
- Prokaryotic gene prediction methods
- Eukaryotic gene prediction methods
- phylogenetic analysis
- Computational Proteomics: Protein visualization tools
- Growth kinetics of industrial strains-includes media design and parameter controls
- Fermentation experiments including product analysis and downstream processing
- Batch and/or chemostat experimental design and implementation
- Bioprocessing of microbial and/or plant (including algal) systems in the context of Biofuels, bioplastics, enzymes and/or other chemicals.
- Field visit and/or miniproject -local bioprocessing unit, Bioreactor design (lab scale vs hungate tube)